Keshav Sharma

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EDUCATION

Vellore Institute of Technology — **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) — *Graduation: May 2027*

SKILL SUMMARY

Data Analytics & Statistics: Statistical Modeling, Predictive Modeling, Time-Series Analysis, Anomaly Detection, Probabilistic Modeling (BG/NBD, Gamma-Gamma), Python, SQL, Pandas, NumPy, Power BI

Optimization & Decision Analytics: Combinatorial Optimization (QUBO), Simulated Annealing, Hybrid Heuristics, Problem Formulation & Trade-off Analysis

Generative AI & LLMs: LLMs, Prompt Engineering, Retrieval-Augmented Generation (RAG), Google Gemini API, AI-Assisted Analysis (Julius AI, GitHub Copilot, Cursor, Claude)

Tools & Platforms: Git, GitHub, Excel, Jupyter Notebook, Streamlit, Docker

EXPERIENCE

VOIS – Vodafone Idea Foundation **|LINK** — ***Aug – Sep, 2025***

- Applied Julius AI and LLM-based prompt engineering to analyze 2 real-world datasets (8,500+ records), extracting 15 structured business insights through natural language querying and AI-assisted data analysis.
- Iteratively refined prompts and validated AI-generated outputs for consistency, accuracy, and relevance, ensuring insights were reliable enough to act on.
- Presented findings to 3 mentors and translated AI-driven analysis into 4 actionable business recommendations on user engagement and content trends, in a client-style stakeholder review.

PROJECTS

Customer Lifetime Value Engine **|LINK**

- Built a probabilistic predictive model using BG/NBD (purchase frequency) and Gamma-Gamma (monetary value) on 1M+ historical transaction records to forecast customer lifetime value.
- Ran the full analytics workflow — data cleaning, feature engineering, model training, validation — then converted outputs into value-based customer segments to support business prioritization decisions.
- Visualized cohort-level trends and revenue potential in Power BI dashboards, translating statistical output into stakeholder-ready insight.

Real-Time Stock Market Anomaly Detector **|LINK**

- Built a statistical and deep-learning anomaly detection pipeline (Isolation Forest + LSTM Autoencoder) on NSE stock data to flag abnormal price/volume behavior.
- Incorporated FinBERT-based sentiment scoring on financial news as a secondary signal, combining structured and unstructured data into one decision framework.
- Evaluated precision/recall trade-offs across models — a direct exercise in balancing false positives against detection sensitivity, the same trade-off decision analytics work requires.

Docwise **|LINK**

- Engineered an AI-powered PDF Question Answering system using Google Gemini, achieving 95%+ query accuracy while cutting manual document search time by 40% via Retrieval-Augmented Generation (RAG).
- Applied prompt engineering and retrieval optimization to reduce hallucinated responses and improve answer reliability.

API Rate Limiter Library **|LINK**

- Implemented and compared five industry-standard throttling algorithms (token bucket, leaky bucket, fixed/sliding window, sliding log), analyzing trade-offs in accuracy, memory usage, and throughput to select optimal strategies for different traffic patterns.
- Designed a modular, storage-agnostic architecture (Redis-backed) enabling algorithm selection based on system constraints — a direct exercise in optimizing for competing resource and performance requirements.
- Validated correctness and edge-case behavior with 30+ automated PyTest unit tests, and published the library on PyPI for public use.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional **|LINK** — 2026
- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate **|LINK** — 2026
- BCG Data Science Job Simulation **|LINK** — 2025